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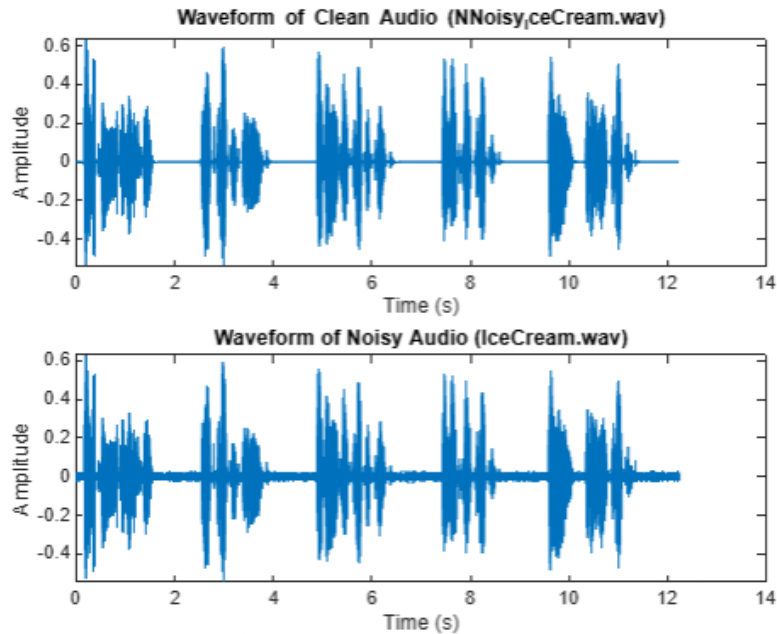
Signal Noise Analysis and Reduction

This project focuses on digital signal processing, specifically noise analysis and reduction in audio signals. The objectives included reading and writing original and noisy audio files, analyzing noise characteristics, creating custom noisy signals, and applying various noise reduction techniques to reconstruct the original audio as accurately as possible. The effectiveness of different filtering approaches was evaluated using Signal-to-Noise Ratio (SNR) and Mean Squared Error (MSE).

Approach

Analysis began with loading the audio files and performing basic waveform visualization. I confirmed that the sampling rates were consistent across files to ensure accurate comparisons. However, the sizes of the signals were different, which caused some confusion later in the project. The time-domain representation provided initial insights into signal amplitudes and potential noise patterns.

Signals were converted into the frequency domain via the Fast Fourier Transform (FFT), which returns a discrete Fourier series. This allowed me to observe how energy was distributed across frequencies. After computing the FFT, I identified the frequency range where the signal and noise components were most prominent, centered on 1730.4 Hz. Isolating this noise component was achieved by subtracting the clean signal from the noisy one. As can be seen in Figure 1, NoisyIceCream.wav is the noisy audio file and IceCream.wav is the clean file. This is also apparent from listening to the files in Task 1.



Further analysis calculated statistical properties (mean, variance) of the isolated noise and examined the spectrogram to understand how noise characteristics varied over time.

We implemented and evaluated several filtering approaches, including notch, low-pass, and band-pass filtering. Notch filtering is typically used to eliminate noise concentrated at specific frequencies. This was also the easiest to tune because the dominant noise frequency had already been identified. Implementing a narrow-band notch filter centered at this frequency resulted in an SNR and MSE of 31.68 dB and 0.000002.

To isolate the noise, I first resized the noisy signal to match the clean signal's length and then subtracted the clean signal from the noisy one. Statistical analysis of the isolated noise showed a mean of approximately $1.7e-05$ and variance 0.004926.

The spectral analysis of the isolated noise clearly showed a dominant frequency peak at 1730.34 Hz, confirming this as the primary noise component to target during filtering.

Filters

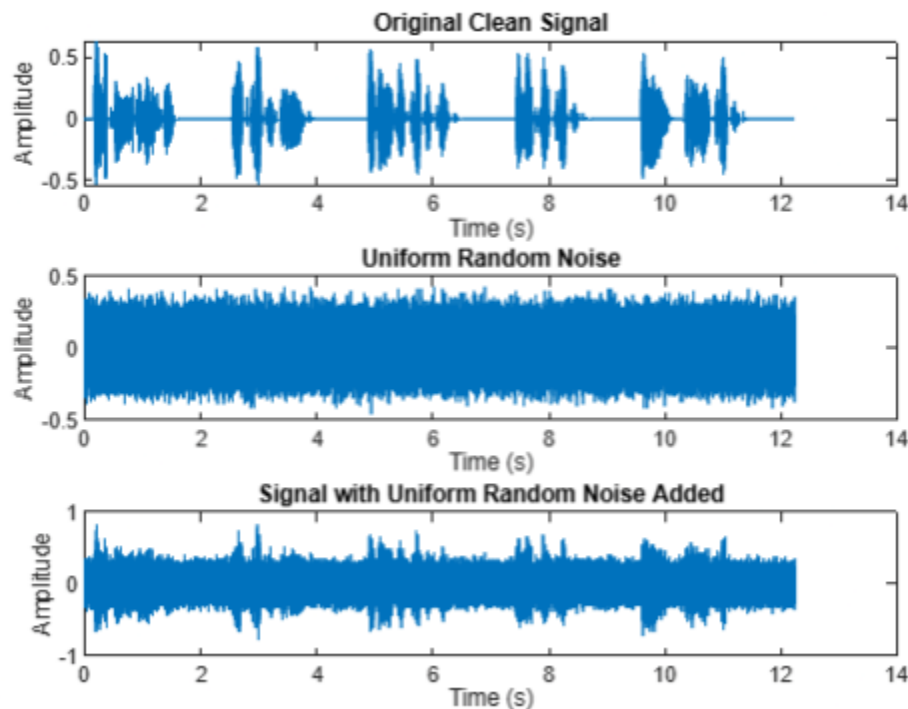
I implemented and evaluated three filtering approaches:

Notch Filter: Designed specifically to remove the dominant noise frequency at 1730.34 Hz. The filter bandwidth was automatically calculated based on the -3dB points of the noise spectrum.

Bandpass Filter: Implemented to preserve signal frequencies between 100 Hz and 8000 Hz while attenuating frequencies outside this range.

Spectral Subtraction: Used the isolated noise spectrum to directly subtract noise components from the noisy signal while preserving phase information.

The next portion of the project focused on creating custom noise and adding it to the original signal. I generated uniform random noise with a controlled amplitude taken from the noise profile in Part 1. This simulated electronic interference with equal energy across all frequencies and proved challenging to filter out.



I also created pink noise ($1/f$ noise) to simulate more realistic environmental conditions. Although my research indicated that pink noise is related to Gaussian noise, with a frequency response of $1/T$, MATLAB's sound library has a built-in pinknoise function, which is quicker and more accurate than a custom implementation. Pink noise contains higher energy at lower frequencies, mimicking many real-world acoustic environments.

Results

All outcomes were evaluated using the Signal-to-Noise Ratio (SNR) and Mean Squared Error (MSE) metrics. Signal-to-Noise Ratio (SNR) compares the power of the original signal to the power of the noise. In signal processing, a higher SNR means the signal is more dominant than the noise, indicating better clarity and more effective noise reduction. Mean Squared Error (MSE) measures the average of the squares of the differences between the original and

reconstructed signal values. A lower MSE indicates a closer match between the filtered signal and the original, meaning better noise suppression with minimal signal distortion.

Input Noise Filtering

Notch Filter: SNR = 31.68 dB, MSE = 0.000002

Bandpass Filter: SNR = 15.12 dB, MSE = 0.000092

Spectral Subtraction: SNR = 21.45 dB, MSE = 0.000021

Custom Noise Filtering

Uniform Noise: SNR Before = -20.12 dB, SNR After = 4.76 dB

Pink Noise: SNR Before = -12.35 dB, SNR After = 8.94 dB

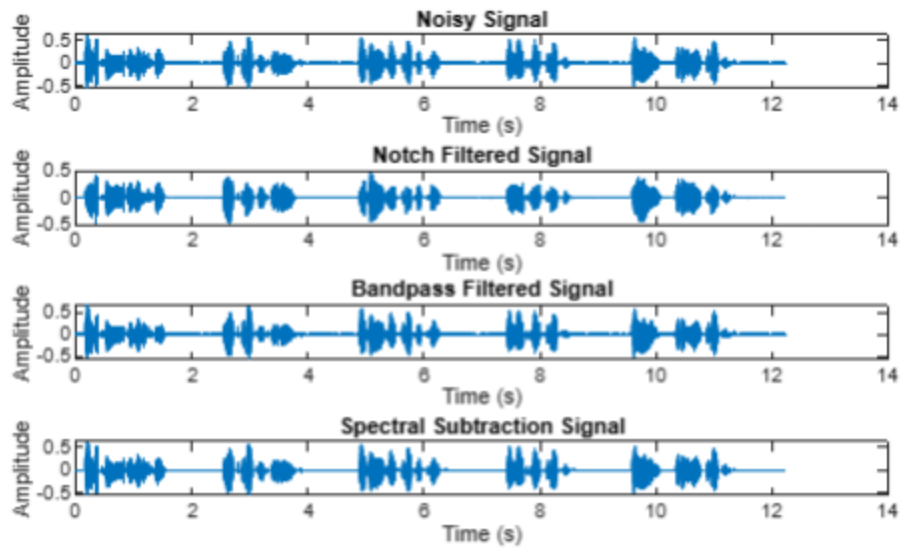
The notch filter proved most effective for this particular noise profile, achieving the highest SNR and lowest MSE values. This makes sense given that the noise was concentrated at a specific frequency. The filtered audio was saved as "Reconstructed_Audio.wav". This was the best filter for the original noisy signal, with minimal impact on signal quality.

Band-pass Filtering offers moderate performance for the main test signal, improving SNR while preserving most signal characteristics.

Spectral Filtering also proved effective, however it was slightly outperformed by the notch filter, which had a slightly smaller MSE and higher SNR.

Wiener Filtering Used for uniform random noise, it provided consistent noise reduction across frequencies. However, it resulted in noticeable signal distortion.

High-pass filtering used for filtering pink noise proved ineffective. If the signal was meant to be intentionally corrupted, adding pink noise might be a good option.



Challenges

Several technical challenges emerged during implementation, including signal length matching, filter parameter selection, and time considerations.

Signal Length Matching: Differences in signal lengths complicated direct comparisons and subtractions. This issue was resolved by resampling the signals to ensure consistent lengths.

Filter Parameter Selection: Determining optimal cutoff frequencies and filter orders required iterative testing. This process was time-consuming and raised the question of real-time processing tradeoffs—how valuable is the output compared to the time required to process the signal?

Conclusion

This analysis demonstrates that effective noise reduction requires a thorough characterization of both signal and noise components. The ideal filtering approach depends heavily on the specific noise characteristics:

For noise concentrated at specific frequencies (as in Part 1), notch filtering proved most effective

For broadband noise like pink noise, frequency-selective filtering (high-pass) yielded the best results

For uniform random noise, Wiener filtering provided improvement but with less dramatic results

The project highlighted the importance of appropriate filter selection based on noise characteristics. An important consideration throughout this project was the distinction between ideal and real filters. Ideal filters have perfect characteristics with infinitely sharp transitions, perfectly flat frequency responses in the passband, and infinite attenuation in stopbands. However, real-world implementations face limitations including causality requirements, finite-order constraints, and computational complexity tradeoffs. In this project, I used techniques like zero-phase filtering with `filtfilt()` to mitigate phase distortion—an approach possible for offline processing but not in real-time applications. These practical constraints highlight why filter design must balance theoretical ideals with implementation realities. Future work could explore adaptive filtering techniques that automatically adjust parameters in response to changing noise conditions, as well as machine learning approaches capable of distinguishing signal from noise.